# Assignment 3

Build Unit Test Automation Solution — Complete Guide

Project: **ONE-CLICK-AGENT**

github.com/3101hasann/ONE-CLICK-AGENT

**Group members**

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Course: Software Quality Engineering

# 1. What Assignment 3 asks for

The assignment is about **automated unit testing**. We take our Python project and write small programs (tests) that call the project's functions with known inputs and check the outputs are correct. All tests run with one command (`pytest`). If anyone later breaks the code, a test fails and warns us.

The marked tasks are:

| Task | Marks | What it means |
|---|---|---|
| Task 1 | 5 | Set up the project locally and record a short setup video. |
| Task 2 | 40 | Pick 3 modules; write 5-10 unit tests per function using ECP/BVA; commit notes. |
| Task 3 | 10 | Put >=4 test cases in Excel and read test data from an external CSV file. |
| Task 4 | 10 | Configure test groups, parallel runs, parameterization and an execution order. |
| Summary | 3 | A summary report before the due date. |
| Distribution | 2 | Git history must show each member's individual work. |

# 2. Team roles and task allocation

Task 1, Task 3, Task 4 and the summary are shared. Task 2 (the big one) is split so each member owns a different module and can be graded individually.

| Member | Owns (Task 2) | Their tests | Also did (shared) |
|---|---|---|---|
| M. Hassan Ahmad (24F-3101) | app/screen_change_detector.py | 27 tests | config tests, CSV/Excel, pytest config, docs |
| M. Ahmad Asghar (24F-3016) | app/response_models.py | 60 tests | config tests, CSV/Excel, pytest config, docs |

**Separate submission folders** (each self-contained) were created in the project:

- `24F-3101_SQE_ASSIGNMENT3/` — Hassan's package (116 tests)
- `24F-3016_SQE_ASSIGNMENT3/` — Ahmad's package (149 tests)

# 3. The three modules under test

## 3.1 app/screen_change_detector.py (Hassan)

Decides whether the screen changed enough to re-send to the AI, so the app never re-uploads an unchanged screen. Functions tested: `compute_signature` (shrinks an image to a 96x64 grayscale fingerprint), `difference` (how much two fingerprints differ), and the `ScreenChangeDetector` class (check / reset / mark_processed). Key boundary: a cell counts as 'changed' only when its brightness moves by more than 24 (strict greater-than).

## 3.2 app/response_models.py (Ahmad)

Validates the structured responses from the AI before the app trusts them. Tested: `parse_model_output` (accepts good JSON, rejects empty/broken input), `FormField.confidence` (a value above 1 is treated as a percentage and clamped to 0-1), and `FeedbackQuestion` coercion for the chosen option index and click coordinates (kept only within 0-1000).

## 3.3 app/config.py (shared)

Loads settings and keeps them within safe limits. The main target is `AgentSettings.clamp()`, which bounds eight numeric settings — ideal for boundary testing (below-min, at-min, middle, at-max, above-max). Also tested: `AppConfig.problem`, `load_config` (reading environment variables) and settings save/load.

## 4. Test-count matrix

| Module / scope | Function area | New tests | Member |
|---|---|---|---|
| screen_change_detector | compute_signature / difference / detector | 27 | Hassan |
| response_models | parse / confidence / coercion / defaults | 60 | Ahmad |
| config | clamp / problem / load_config / settings I/O | 69 | Shared |
| CSV-driven | 4 data-driven functions | 20 | Shared |
| New total | | 176 | |
| Existing regression | preserved, unchanged | 39 | pre-existing |
| GRAND TOTAL | | 215 | |

Each parametrized case counts as one test (this is how pytest reports them). Every selected function has at least 5 tests, meeting the 5-10 requirement.

## 5. Testing techniques used

- **ECP (Equivalence Class Partitioning)** — one example from each category of input (valid, invalid, missing).
- **BVA (Boundary Value Analysis)** — the edges: e.g. clamp ranges tested at below-min, min, mid, max, above-max; cell_delta at 23/24/25.
- **Negative / exception testing** — broken JSON, wrong types, out-of-range values.
- **Decision testing** — the config 'problem' table (mock / no key / no model / ok).
- **Parameterization** — one test function runs many input cases.
- **Regression** — the original 39 tests are kept and still pass.

## 6. Task 3 — Excel and CSV

**CSV:** `tests/data/assignment3_test_data.csv` holds 18 rows of test inputs. Four test functions read this file at run time and turn each row into a test (config clamp, confidence, chosen_index, and the difference boundary).

**Excel:** `docs/assignment3/test_cases.xlsx` lists 18 documented cases with columns (ID, Module, Function, Scenario, Technique, Input, Steps, Expected, Actual, Status, Member). It is generated from the same CSV so every ID traces 1:1.

## 7. Task 4 — Groups, parallel and order

Configured in `pytest.ini` and `conftest.py`.

- **Groups (markers):** smoke, config, validation, screen_detection, csv_data, regression.
- **Parallel:** `pytest -n auto` (pytest-xdist).
- **Parameterization:** used across all files.
- **Order:** smoke -> config -> validation -> screen_detection -> csv_data -> regression (sequential runs; parallel may interleave, which is fine because tests are independent).

## 8. Setup commands (one time)

```
cd C:\Users\Hassan\cheater-exe
python -m venv .venv
```

```
.venv\Scripts\Activate.ps1
pip install -r requirements.txt
pip install -r requirements-dev.txt
```

If activation is blocked: Set-ExecutionPolicy -Scope CurrentUser RemoteSigned. Tests need no API key or internet.

# 9. Running the tests

## Everything / assignment / parallel

```
python -m pytest                 # whole project  -> 215 passed
python -m pytest -m assignment3  # our new tests   -> 176 passed
python -m pytest -n auto         # parallel        -> 215 passed
python -m pytest -v              # show every test name
```

## Only MY work (Hassan, 24F-3101)

```
python -m pytest -m screen_detection   # my 27 tests (individual Task 2)
cd 24F-3101_SQE_ASSIGNMENT3 & ..\.venv\Scripts\python.exe -m pytest   # my folder -> 116
```

## Only AHMAD's work (24F-3016)

```
python -m pytest -m validation         # his 60 tests (individual Task 2)
cd 24F-3016_SQE_ASSIGNMENT3 & ..\.venv\Scripts\python.exe -m pytest   # his folder -> 149
```

## By group / by file / CSV

```
python -m pytest -m config            # 69
python -m pytest -m csv_data -v       # 20 (reads the CSV)
python -m pytest tests/unit/test_config_assignment3.py
python -m pytest --collect-only -q    # see the run order
```

# 10. Folder layout

```
cheater-exe/
  app/                       the application being tested
  tests/                     live test suite (runs the whole project)
    test_*.py                existing regression tests (39)
    unit/                    new assignment tests (config, validation, screen)
    data/                    CSV + CSV-driven tests
  pytest.ini, conftest.py    markers, parallel, ordering config
  requirements-dev.txt       pytest-xdist, openpyxl
  docs/assignment3/          all docs, Excel, this guide
  24F-3101_SQE_ASSIGNMENT3/ Hassan's submission package (tests, docs, video)
  24F-3016_SQE_ASSIGNMENT3/ Ahmad's submission package (tests, docs, video)
```

# 11. Verified results

| Command | Result |
|---|---|
| python -m pytest (full, sequential) | 215 passed |
| python -m pytest -n auto (parallel) | 215 passed |
| python -m pytest -m assignment3 | 176 passed |
| Hassan folder (24F-3101) | 116 passed |
| Ahmad folder (24F-3016) | 149 passed |
| python -m pytest -m csv_data | 20 passed (reads CSV) |

No failures, no skips, no errors. Run in the project venv on Windows, Python 3.14.6.

# 12. What is still to be done (by the group)

- Each member commits their own files under their own Git identity (see member_contributions.md) so the history shows individual work.
- Push to GitHub when ready, make the repository public, and submit the link on GCR.
- Setup videos are included (terminal recordings of the real test runs); re-record as a desktop screencast if the evaluator requires it (script in setup_video_script.md).
- The repository has no LICENSE file (left as-is by decision); noted in the docs.

## 13. Deliverables checklist

| Deliverable | File(s) | Status |
|---|---|---|
| Setup guide | docs/.../setup_guide.md | Done |
| Setup video + script | video/*.mp4, setup_video_script.md | Done |
| Unit tests (3 modules) | tests/unit/*.py | Done (176) |
| Test inventory + matrix | test_inventory.md | Done |
| ECP/BVA design notes | test_design_notes.md | Done |
| Excel workbook | test_cases.xlsx | Done (18) |
| External CSV + driven tests | tests/data/* | Done (4 funcs) |
| Pytest config (groups/parallel/order) | pytest.ini, conftest.py | Done |
| Execution guide | test_execution_guide.md | Done |
| Member contributions | member_contributions.md | Done |
| Summary report (md + Word) | summary_report.md/.docx | Done |
| This complete guide | Assignment3_Complete_Guide.pdf | Done |